ABSTRACT OF THE DISCLOSURE

A covert illumination and imaging apparatus indistinguishable from a standard consumer light emitting diode (LED) bulb. The apparatus comprises a curved, flexible polymer substrate conforming to the interior geometry of a translucent bulb enclosure. The substrate supports an interleaved array of self-emissive subpixels (e.g., OLED or QDEL) and micro-scale organic photodetector (OPD) sensor subpixels. Driven by a de-phased duty cycle, the light-emitting subpixels provide room illumination matching the standardized lumen output of a conventional bulb. During sub-millisecond dark intervals, the integrated OPD sensors capture wide-angle, omnidirectional images of the surrounding environment. The system enables undetectable covert surveillance without the use of discrete, visually detectable camera modules or physical apertures.